The edge of the $3m$ wide truck corresponds to $x = 1.5m$ from centre We will find the height of the archway $1.5m$ from the centre by substituting $x = 1.5$ and solving for y

$$\frac{\left(\frac{3}{2}\right)^2}{36} + \frac{y^2}{9} = 1$$

$$y^2 = 9\left(1 - \frac{9}{144}\right)$$

$$= \frac{9(135)}{144} = \frac{135}{16}$$

$$y = \frac{\sqrt{135}}{4}$$

$$= \frac{11.62}{4}$$

$$= 2.90$$

Thus the height of arch way $1.5m$ from the centre is approximately $2.90m$. Since the truck's height is $2.7m$, the truck will clear the archway.

Example 5.32

The maximum and minimum distances of the Earth from the Sun respectively are 152×10^6 km and 94.5×10^6 km. The Sun is at one focus of the elliptical orbit. Find the distance from the Sun to the other focus.

Solution

$$AS = 94.5 \times 10^6 \text{ km}, \; SA' = 152 \times 10^6 \text{ km}$$

$$a + c = 152 \times 10^6$$

$$a - c = 94.5 \times 10^6$$

Subtracting $2c = 57.5 \times 10^6 = 575 \times 10^5$ km

Distance of the Sun from the other focus is $SS' = 575 \times 10^5$ km.

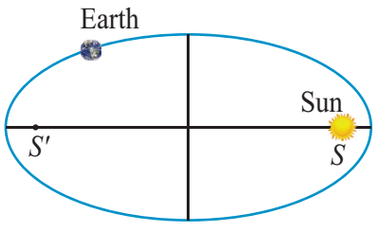


Fig. 5.56

Example 5.33

A concrete bridge is designed as a parabolic arch. The road over bridge is $40m$ long and the maximum height of the arch is $15m$. Write the equation of the parabolic arch.

Solution

From the graph the vertex is at $(0,0)$ and the parabola is open down

Equation of the parabola is $x^2 = -4ay$

$(-20,-15)$ and $(20,-15)$ lie on the parabola

$$20^2 = -4a(-15)$$

$$4a = \frac{400}{15}$$

$$x^2 = \frac{-80}{3} \times y$$

Therefore equation is $3x^2 = -80y$

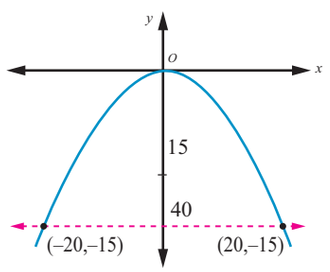


Fig. 5.57

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